

Unit IV

Product Analysis and Strategic Development

Product teardown process and methods, Benchmarking approach and detailed procedure, Tools used in benchmarking, Trend analysis and setting product specifications, Product portfolio and architecture, Function-form diagrams, Applications of reverse engineering in product development.

Reverse Engineering

A reverse engineering and redesign methodology begins with the current product.

- One of the first tasks is to understand the market for the current product.
- A customer needs analysis is required, focusing on what customers like and what they don't.
- Based on this, a variety of redesign opportunities will become apparent.
- A key step in reverse engineering is to make intelligent estimates as to what the functional model ought to be using the modelling.
- The next step is to dissect the product and understand how it operates to satisfy or not satisfy customers.
- Based on the tear down of several products in the marketplace and related products of the manufacturer, the product can be analyzed.
- Understanding the component and the system technology of the company and its competitors a real function structure for the product can be developed.

Product Tear Down Process

- Generally, teardowns are carried out in industry to Benchmark against the competition.
- They observe the technology and architecture and undercover principles behind it works.
- A traditional product teardown involves far more than disassembly of the product to see how it is put together. One must analyze the system and transform this analysis into information that can be used as a part of new design.

Step 1: List the design issues

This first step must be clear what problems and opportunities the design team is facing on the current project.

- If it is a new project, the design issues may be unknown, and every fact about customer, market competitors, and features of competitors' products are worth investigating.

- If the project entails a redesign, an investigation can ask the design team:
 - What was difficult for them?
 - What design problem did they solve that they are proud of?
 - What related technologies were they interested in?
- The last set of design issues that need recording is basic information on the component in assemblies. This information needs to evaluate the performance criteria in the latest analysis.
- Material types and masses are needed to determine part cost, which are typical basic factors that need to be known in industrial design process. These include:
 - Quantity of parts per product unit.
 - Dimensional measurement.
 - Max, min, and average material thickness.
 - Weight, material, color/finish.
 - Manufacturing process.
 - Cost per part/subassembly.

Step 2: Prepare for product tear down

After identifying the design issues, one should identify all tools that will be required to complete the tear down. This activity includes identifying all sensors and test equipment required for the measurement process. Typical equipment includes:

- Camera (?), videotape (of product operating?), optical sensors (?).
- Flow meters (?), Dynamometers (?), Calipers (?).
- Typically, one documents their information into a written or electronic report of the teardown.

Step 3: Examine the distribution and installations

- Important factors in the product development decision-making process that must not be overlooked are the means used to acquire parts, contain them, shipment, distribute, and market the product.
- These must also be examined as part of a teardown process.
- The distribution packaging of a product should be examined and reported to the design team. It can often be quite expensive.
- Consumer installation instructions and procedures should be examined for costs, effectiveness, and liability.

Step 4: Dis-assemble, measure and analyse data by assemblies

- Disassembly is the obvious step commonly pictured when thinking of reverse engineering.
- This step must be coordinated with measurements and experimentation.
- First take pictures and measurements on the whole assembly before disassembly.

- Take apart the assembly.
- Take a picture in an exploded view (or produce a solid model assembly diagram- electronic exploded view).
- Take measurements on the parts and assemblies to complete the data sheet.
- In this test, avoid destructive test.
- Parts that are fabricated with insert, molds, rivets, welds, solder (such as windings on motor and mechanical springs in mechanical clocks) should be carefully disassembled so that the product can still function.

Step 5: Form a bill of material

- During disassembly, the team should complete a written form that details the product.
- Assembly photos and exploded view CAD drawing should be scheduled.
- The data collected in each column of the BOM is required for analysis including cost and performance.

Material List					
Part	Part Name	Quantity	Functions	Mass	DDM cost analysis data

Teardown Methods

a) Subtract and operate

The subtract and operate method involves five steps aimed at expanding redundant components in an assembly or sub-assembly through the identification of the true functionality of each component (Leatver and wood 1996).

Step 1: Dissemble (subtract) one component of the assembly, removal of component may occur in any order.

- The team may be required to remove one or several components in order to remove the desired component.
- These pre-requisite components should be reassembled if possible.
- If they can't be reassembled, measures should be used to replicate the function of the missing components.

Step 2: Operate the system through it's full range

- This step should test the product through the range of customer needs.
- After removing a component, the product should be thoroughly tested.
- For each customer need (ergonomics, kinematics, etc.), the product must be tested to verify the effect due to the removal of the component.

Step 3: Analyse the effect

- This step is most commonly used through a visual analysis.
- However, sometimes testing devices are used.

Step 4: Deduce the sub function of missing component

- From step (3) the function of the missing component may be deduced.
- A change in any degree of freedom (DOF) during operation should be a major focus.
- DOF is a critical issue in determining component redundancies.

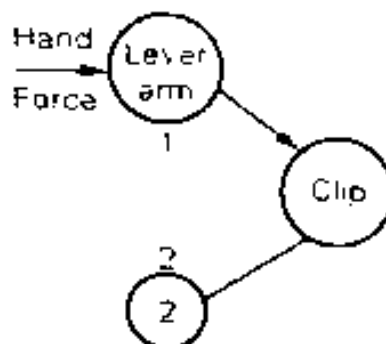
Step 5: Replace the component and repeat the procedure

- Repeating the procedure '**n**' times, where '**n**' is the number of pieces in the assembly.

Force Flow Diagrams/Energy Flow Diagrams

Force flow diagrams are a method for exposing component functionality and opportunity for component elimination.

- Force flow diagrams focus on component combination.
- A force flow diagram represents the transfer of force through a product's components.
- The components are symbolized as nodes using circles or squares, and forces are drawn as an arrow connecting the components in which the force transfer takes place.
 - *Example:* A Paper clip.
- In the case of the hand transfer of force to each of the lever arms.
- These lever arms in turn transfer the force to the clip.
- The motivation behind constructing a force flow diagram is to map the force flow through a product so that the diagram can be analyzed to help expose opportunities for components/piece combination.



- Once model is formulated the first step in analysing a force flow diagram is to place an 'R' on the flows that have relative motion between two components.

- Once this step is completed the diagram can be decomposed into group separated by R's.
- Having performed this separations the following proposition is applied.

Proposition The components comprising 'R' groups are candidates for combinations, if not prohibited by material or disassembly issues. Combination between member of one group and member of outside of the group requires more complex redesign.

Support/ Explanation Two parts move with respect to each other in a product to provide a function. Eliminating movements means eliminating a function and redesigning a product functional structure is difficult, requiring an additional subsequent redesign of the concept implementing the new function structure.

Steps

Eight steps comprises he process for executing a force flow analysis.

- Step 1: Identify the primary force flows (energy flows) transmitting through a product.
- Step 2: Map the force (energy) flow from external source through each component of product, until the flow exists the ground. Special care should be exercised to identify when force (energy) flows split (become parallel) through components.
- Step 3: Document the result in force flow diagram, where the nodes are components and the connections are the forces.
- Step 4: Analyse the diagram, labelling relative motion between components with an 'R'.
- Step 5: Decompose the diagram into groups separated by 'R's.
- Step 6: Deduce the sub-function and affected customer needs for each group.
- Step 7: Develop a creative conceptual design to combine the components in each group.
- Step 8: Repeat for each force (energy flow).

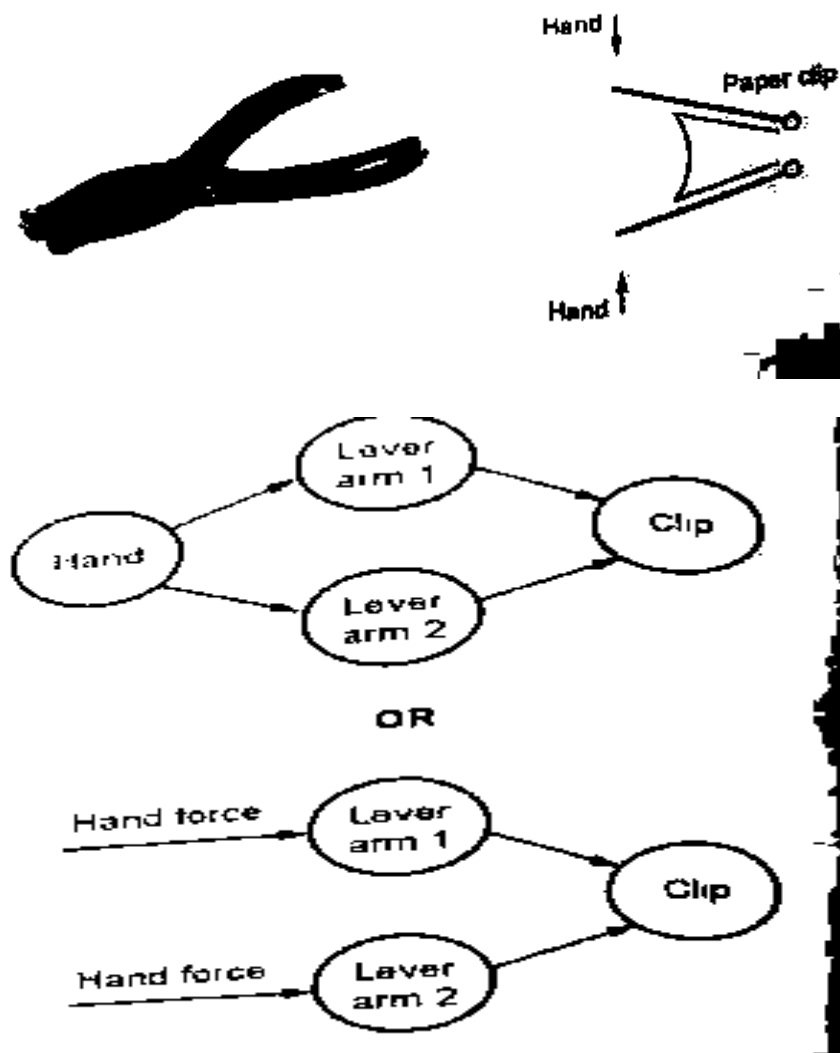
According to Boothroyd (1980) a component in a product must exist if:

1. The parts move relative to other parts.
2. The parts must be made of different material than other parts.
3. Assembly and disassembly of other required parts would be impossible if the part is not separate.
 - Force flow diagram takes this notion a step further by actually showing which components can be combined for product as a system.
 - Modelling the force transfer through the components allows the complete product to be analysed for piece combination.

- Grouping the products according to the criteria (1) [relative motion] highlights the sets of components that can potentially be combined for low level adaptive design.
- The components of the group should be then evaluated with the criteria (2) and (3).

Force Flow Diagram: Paper Punch

- In this case, the hand transfers the force to each of the lever arms. These lever arms in turn transfers the force to the clip.
- Fig. represents the transfer through the force flow diagram.



Measurement and Experimentation

Need for measurement in product design:

- Measurement in engineering design comes in several forms, notably specifications and benchmarking.
- 1. A product specifications provide a precise and measurable description of what product must accomplish.
- The second approach is called benchmarking.
- The importance of measurement in design is illustrated by noticing... how design methods are concerned with ensuring that both the customer needs and functionality of the product are quantifiable.
- Miles states (1971) scores in relevance to determining the functionality of products in value engineering.
- Measurement has evolved quite dramatically over the past 100 years and is a very broad field of study.
- Each specification consists of a metric denoting the types of measurement and target value in the form of number or range.
- The product's target value is the performance level that the designer must achieve.
- This target value is generally determined by examining the usage of the product or by examining the performance of other similar product.

Advance method: Product Development Forensics

- The word forensic is defined in part as pertaining to connected with or used in courts of law or public discussions and debates.
- Forensic and product teardown are similar in that both have a common goal of obtaining the information about the subject using systematic procedure.

Anthropology Forensics

- It deals with the identification of skeletonized remains known to be or suspected of being.
- Three issues are discussed in concerning anthropology forensic that relate to reverse engineering and product teardown.
 1. The procedure used in forensic anthropology and reverse engineering process are very similar and show the importance of common steps.
 2. Secondly the importance and procedure from removing bias during extraction of data.
 3. Finally concept of taking different sets of measurements to evaluate distinct features of the subject can be adopted in reverse engineering process.

Applications of Product Teardown

- Product teardown is helpful to understand the engineering manifested in current product.
- It also exposes creative avenues for product evaluation.
- There are widely used two product applications, where product teardown is widely used. a) The first focuses on the use of teardown, including force flow diagram. The result from these tools, when combined with engineering analysis, leads to active product or product function reduction. b) The second application considers the development of metrics and measurement only for any product. The end result is a set of metrics that addresses the customer needs of the product and corresponding transducers for quantifying these matrices from the product and competitive benchmarks.

Benchmarking Approach

Step 1 : Form a list of design issues

- A list of design issues must be developed for comparative benchmarking.
- Further this list should be continually revised and updated.
- The result is reduction in wasted time and resources.

Step 2 : Form a list of competitive and related products

- For product one must list all related competitor's product and different models.
- This step should only be an identification of competitors, basically company name and product names.
- With complete set of different products, vendors and suppliers to examine, the list should be screened by highlighting the particular competitors that appears more crucial for design team to fully understand.

Step 3 : Conduct information search

- A design team should gather information on
 1. The products and related products
 2. The functions they perform
 3. The targeted market segment
- Written sources of product information :

i) Libraries : Universal libraries are filled with technical engineering modelling references.

ii) Thomos register of companies : This is a set of documents is a "yellow pages" for manufacturing related business. The Thomos Register list vendors by product name. <http://www.thomoregister.com>

iii) Market share reporter : Published every year international, Thomson Publisher. This book includes corporate market share, institutional shares and brand market shares.

iv) National Buereu of standards : This U.S. government branch provides among other thing, national labour rates for all major countries. This information proves very useful for determining competitors manufacturing costs.

v) Census of manufacturer : Taken every five years by U.S. department of commerce provides current industrial reports, list production and shipment data on industries and same products.

vi) Mood's industry review : This review is conducted every six months, this survey provides key financial information, operating data of about 3500 companies.

vii) Consumer reports magazine : This magazine surveys and tests a number of common consumer products, useful data is available for customer needs, qualitative benchmarking, engineering specifications and warranty information.

viii) Trade magazine : Consumers trade magazine such as car and driver, Auto gear, provides competitive studies of product.

ix) Patents : After examining trade journals and uncovering which competitors have bragged about new innovation, gathering patents on these new innovations explain much on [http : // www.patents.ibm.com](http://www.patents.ibm.com) **Other sources** : www, vendors, technical specialists, experts etc.

Step 4 : Teardown multiple products in class

- The result of this step should be an intended bill of materials for each product a functional model for each product an exploded view of each product.

Step 5 : Benchmarking by function

- After benchmarking several comparable products, the team needs a tangible outcome of the analysis.
- One approach is the summarize the comparison by the product form.
- The problem with this approach is that any component in one product may not be functionally equivalent to the same or similar component in other product.

Step 6 : Establish best in class competitors by function

- Having listed various solutions used for every function, comparative analysis can be completed.
- For each function, the highest performing solution can be called out as well.

Step 7 : Plot industry trends

- This process should include

i) Categorization of the market : This is an approach, when one categorizes products solution by socio economic status of the typical customer of individual product and also by percentage of market.

Here's the extracted data from this page:

ii) Benchmarking by technical solution : This is an approach to comparing how products perform. The most effective approach is a development of thorough technology timeline using 's' curve.

iii) The benchmarking of competitors : The thorough study of competitors profile and business is identified for benchmarking.

Tools used in Benchmarking

Intended Assembly Cost Analysis

- One key result of benchmarking activity is comparative understanding of the cost structure that different competitors face.
- Part manufacturing in the U.S. with assembly in China has different cost structure than part manufacturing in Singapore with assembly in Japan.
- These different scenarios call for different design configuration and what may appear a bad design may not be given production scenario.
- Assembly in low cost labour countries can enable the use of complex assembly processes with many inexpensive parts, whose assembly in high labor cost countries requires design with complex multifunctional parts.
- To uncover such understanding one must analyse the part/assembly structure during the teardown and record the assembly structure.
- What is attached to what ?
- Then to each part one can estimate part cost and to each assembly step, one can estimate an assembly cost.
- Generally each of these cost is a function of material, equipment and labour cost.

- One can estimate these cost and aggregate them into an intended cost.
- An intended cost estimate organises the resulting analysis according to the hierarchy of assemblies, sub assemblies, parts and factors.

Function Form Diagrams

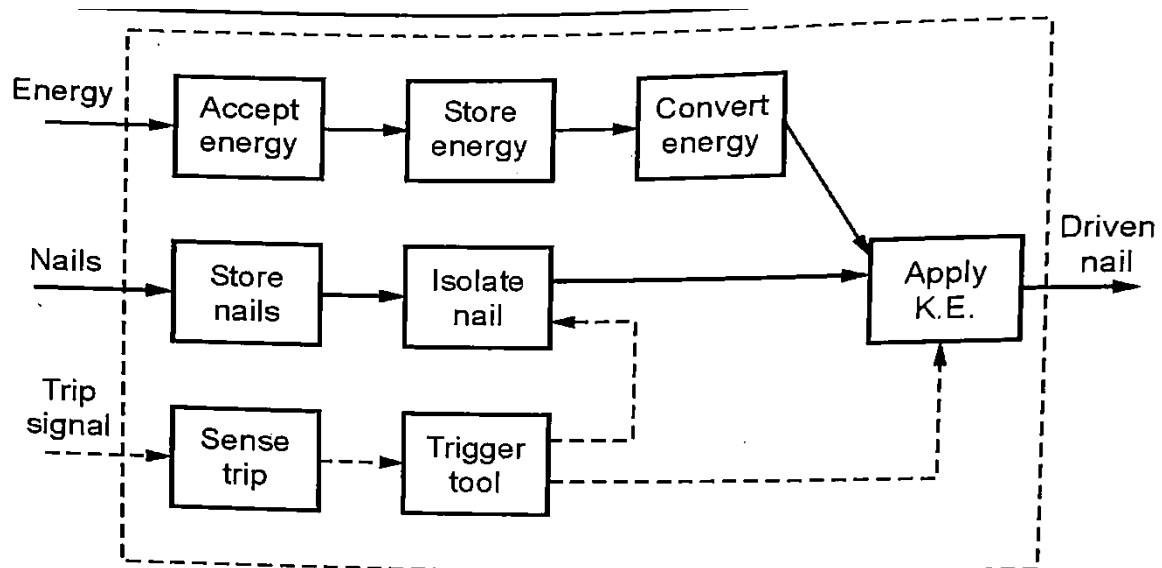
- After analysing the cost of several comparable products one can develop an outcome by comparing cost.
- The result of benchmarking analysis should be listed on the function form diagram such as part cost and material information.
- Finally, the best in class solutions on the market should be called out such as quality and lowest cost models.

Example of Function Form Diagram : Pneumatic Nail Gun

- Fig. shows a typical pneumatic nail gun. These guns use compressed air to drive nails into a variety of materials.
- Therefore if we were constructing a black box diagram for the product the input flows would be compressed air, nails and trip signal.
- Compressed air is a form of energy, nails are a material and the trip signal is information. The products overall function is "Drive Nails".
- The desired flow from the product is a driven nail. Unfortunately there are some other, undesirable flows out of the product.
- These undesirable flows include noise and heat. As designers we may want to minimize these undesirable outputs. The full black box diagram is shown in Fig.



Pneumatic nail gun



The final FSD for the nailer

Trend Analysis

- Final outcomes of the benchmarking analysis are observations that the benchmarking group can make about where the products and industry appears to be headed.
- These observations can be quantitative such as plotting 's' curve.
- Or can also be qualitative such as uncovered comments by senior executives on where they want to place their company in the market.

Setting Product Specification

- Having benchmarked competitive products on customer and technical criteria, one next step is to use this information to set targets for a new product development effort.
- The benchmarking process allows us to understand where there are potential openings in the market.

Specification Process

- Specification for a new product are quantitative and measurable, criteria that the product should be designed to satisfy.
- There are two aspects to a specification that need to be classified.
 - i) The specification is on dimensions that can support the units like meter, degrees, horse power and so forth.
 - A quantity that has units will also be called engineering requirement.
 - In addition for having units through a specification needs a target value.

- This is a number along dimensional unit that establishes required performance.
- Each specification should be measurable.
- Testable or verifiable at each stage of the development process not just at the end of process, when the product is designed and built.
- This stage is a point in the development process, when overall product specifications should be well considered.
- We develop specifications using two approaches.

ii) From a check list viewpoint and

iii) From a viewpoint of the translation of qualitative customer needs.

Product Architecture

- Depending on the target customer population, as well as other factors such as ease of manufacturing and simplicity of design, a product architecture may be configured in various ways. For any given product, product architecture as the strategy by which function is mapped to form.
- There are two main categories of product architecture, integral and modular, which are defined by

(a) The relationship between functions and components, and

(b) The interaction between components.

- An integral architecture implies complex mapping between components and functions and a high level of incidental interaction between components.
- A modular architecture exhibits one-to-one mapping between functions and components and only necessary interactions between components.

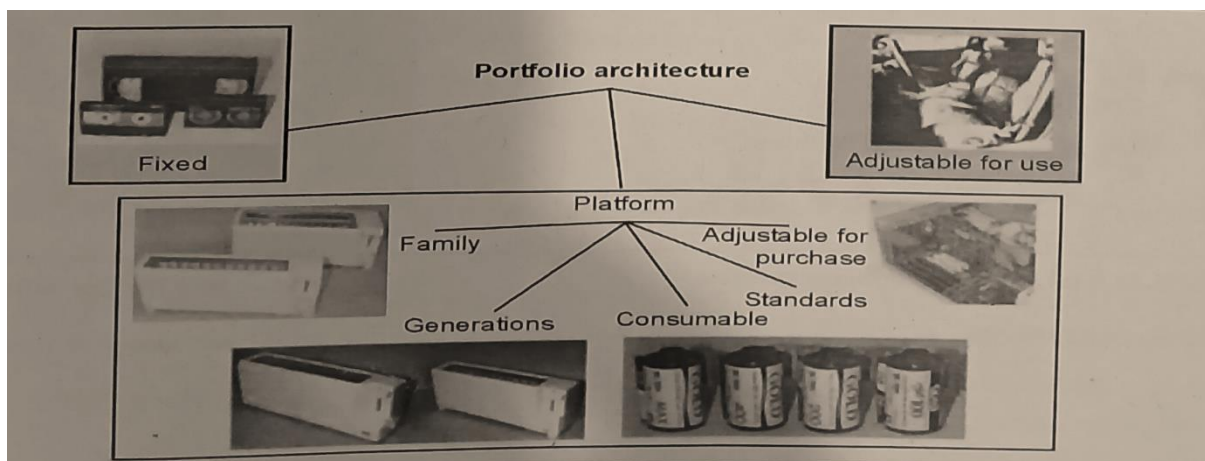
Product Portfolio Architecture

- Although there are advantages to modularity when applied to a single product such as ease of design and facilitation of production, especially with outsourced components, modularity can be leveraged to much greater benefit when implemented in a set of multiple products that draw on common resources.
- **Fixed Portfolio Architecture** : A set of products exhibiting fixed portfolio architecture for a specific feature offers a single option across the entire set. For example, Henry Ford's famously limited original line of automobiles demonstrated a fixed portfolio architecture for color in that black was the only option offered to customers.

- Offering one feature option may be an advantageous decision when there is limited variety in customer demands, a firm has a monopoly, or multiple options are prohibitively expensive to offer.

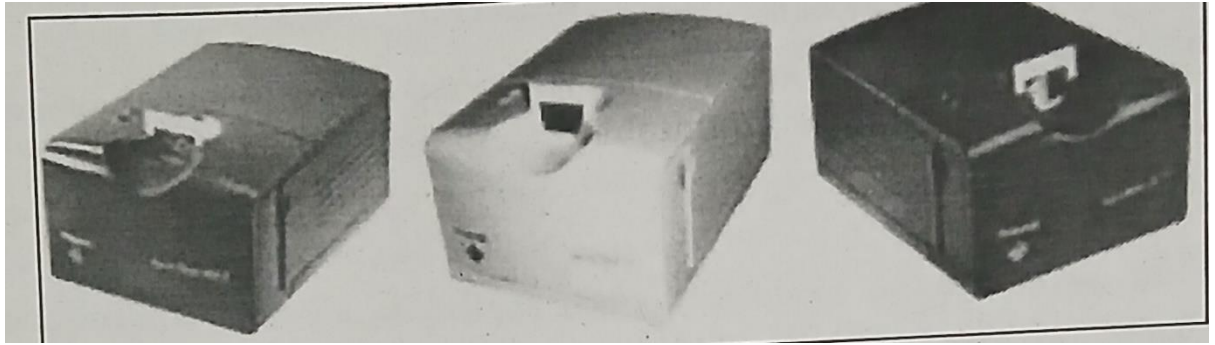
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Various types of portfolio architecture

- **Platform Portfolio Architecture** : On the other hand, a set of products with platform portfolio architecture for a specific feature offers variety through multiple options across the set.
- Usually this implies each product variant in the portfolio exhibits some form of modular product architecture.
- One of the main advantages of modular product architecture lies in the ability to remove or replace modules without affecting the rest of the design, thereby creating different products with minimal investment of time and resources.
- Polaroid's family of photographic slide scanners, for instance, offer three different level of scan resolution.

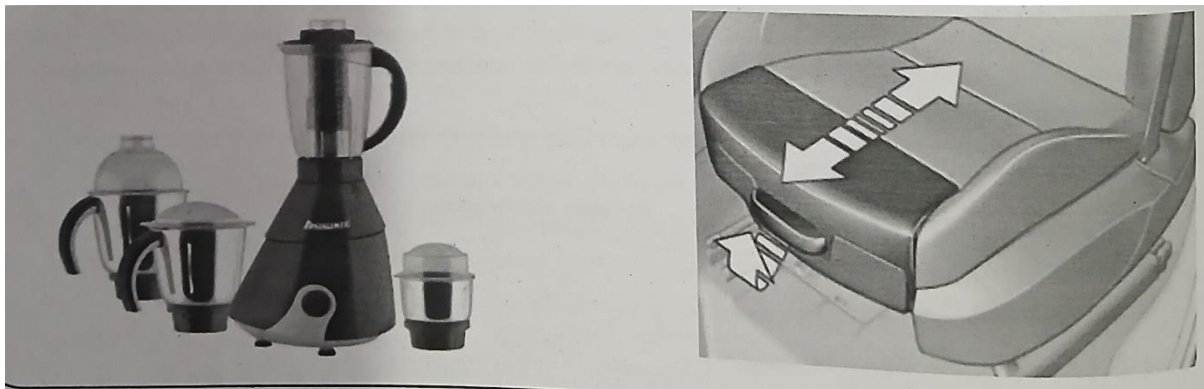


A platform family of polaroid scanners with varying levels of resolution

- **Adjustable Portfolio Architecture** : Lastly, adjustable portfolio architecture offers multiple options through a single design which can be customized by the user.
- The customer can change the value of an adjustable feature at any time, unlike a fixed or platform feature.

(a) Household blenders with one (fixed) or multiple (adjustable) settings for motor speed demonstrate the difference between products with fixed and adjustable features.

(b) Adjustable automobile seats also enable several different feature values for leg room to accommodate drivers of different heights.



(a) Household mixer grinder

(b) Car adjustable seat

Rapid Prototyping

- A unique fabrication processes developed to make engineering prototypes in minimum lead time based on a CAD model of the item.
- The traditional method is machining - Machining can require significant lead-times - several weeks, depending on part complexity and difficulty in ordering materials.
- It allows a part to be made in hours or days given that a computer model of the part has been generated on a CAD system

- Because product designers would like to have a physical model of a new part or product design rather than just a computer model or line drawing - Creating a prototype is an integral step in design -
- A virtual prototype (a computer model of the part design on a CAD system) may not be sufficient for the designer to visualize the part adequately - Using RP to make the prototype, the designer can visually examine and physically feel the part and assess its merits and shortcomings
- There are two basic types of Rapid Prototyping (RP)

1. Material removal RP - machining, primarily milling and drilling, using a dedicated CNC machine that is available to the design department on short notice - Starting material is often wax, which is easy to machine and can be melted and re-solidified - The CNC machines are often small - Called desktop milling or desktop machining.

2. Material addition RP - adds layers of material one at a time to build the solid part from bottom to top

Material Addition RP

1. Liquid(s) that are cured layer by layer into solid
2. Powder(s) that are aggregated and bonded layer by layer to form a solid
3. Solid sheets or filaments that are laminated to create the solid part

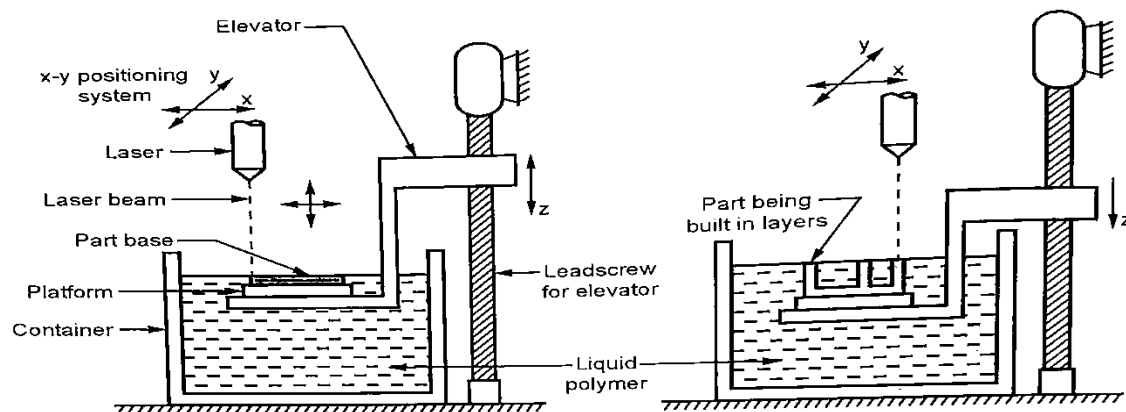
Classification of Rapid Prototyping Technologies

- There are various ways to classify the RP techniques that have currently been developed. The RP classification used here is based on the form of the starting material :
- 1. Liquid-based 2. Solid-based 3. Powder-based

Liquid-Based Rapid Prototyping Systems

- Starting material is a liquid. About a dozen RP technologies are in this category can be listed as follow,
- 1. Stereolithography 2. Solid ground curing 3. Droplet deposition manufacturing
- Stereolithography (STL/SLA) RP process for fabricating a solid plastic part out of a photosensitive liquid polymer using a directed laser beam to solidify the polymer.
- Part fabrication is accomplished as a series of layers, in which one layer is added onto the previous layer to gradually build the desired 3-D geometry.

- The first addition RP technology - introduced 1988 by 3D Systems Inc. based on the work of Charles Hull
- More installations of STL than any other RP method.
- Fig. Stereolithography : (a) At the start of the process, in which the initial layer is added to the platform; and (b) After several layers have been added so that the part geometry gradually takes form (See Fig. on next page)



Questions

1. What is product teardown ? Explain the teardown process in details.
2. Discuss measurement and experimentation in product design.
3. What are the different applications of product teardown.
4. Explain the stepwise procedure of benchmarking in details. Explain the function form diagram with example.
5. What are different tools used in benchmarking.
6. Explain in details the process of setting product specifications with suitable example.
7. Explain the product portfolio and architecture.